GIT MODULE · Week 9 · Mon

Git & Version Control

Track changes. Collaborate fearlessly. Never lose work.

K0126002 · Gachon University · 2026 Fall · Dr. Toshpulatov

What is Version Control?



Track Changes

Every save is a snapshot. See exactly what changed, when, and who changed it.



Rollback Anytime

Broke something? Rewind to any previous working state in seconds.



Collaborate Safely

Multiple people edit the same project without overwriting each other.

Git vs GitHub

git

Git

- Free, open-source tool
- Runs on YOUR computer
- Tracks changes locally
- Works offline

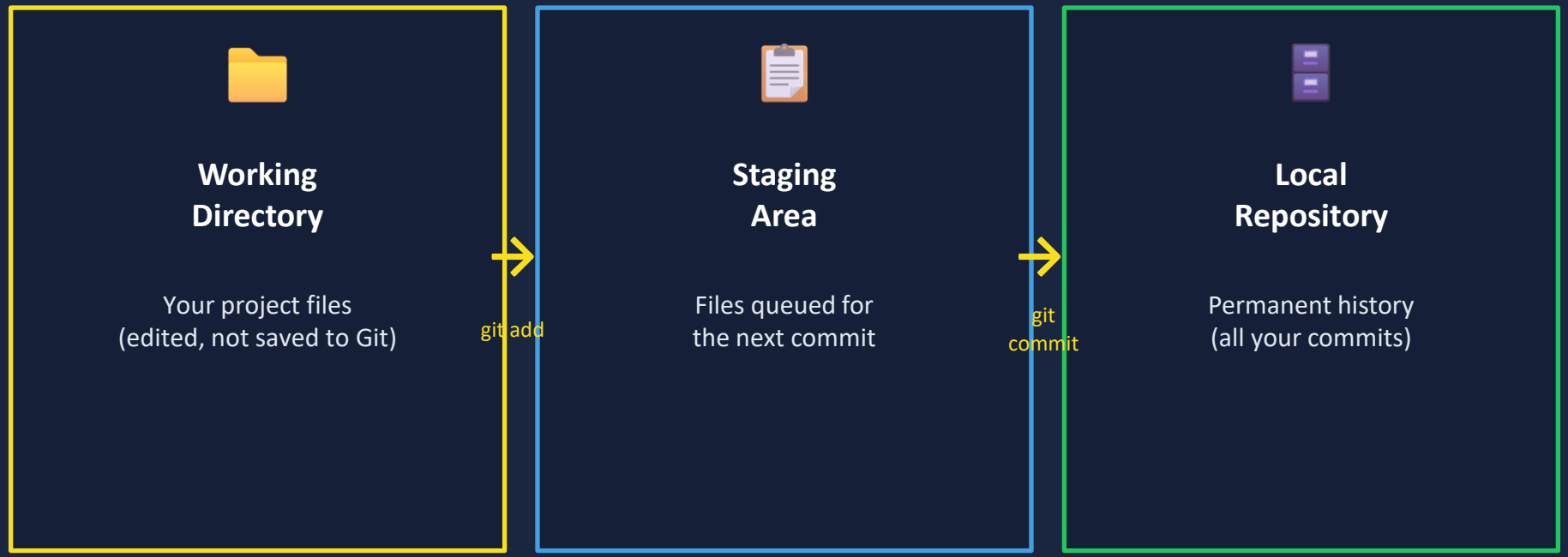
GH

GitHub

- Cloud hosting platform
- Stores your Git repos online
- Team collaboration hub
- Free for public repos

+ Analogy: Git = MS Word Track Changes (local) | GitHub = Google Drive (cloud sharing)

The Git Workflow



Remote Repository (GitHub) ← git push / git pull → teammates

QUICK QUIZ

What does `git init` do?

A Installs Git on your computer

B Creates a new Git repository in the current folder

C Clones a remote repository

D Commits all current files

Discuss — then we'll reveal the answer

✅ ANSWER

What does `git init` do?

A Installs Git on your computer

B Creates a new Git repository in the current folder

C Clones a remote repository

D Commits all current files

git init creates a hidden .git/ folder that starts tracking your project. It does NOT clone, push, or install anything.

Core Commands — Start & Save

```
# 1. Create a new local repository
$ git init my-project
$ cd my-project

# 2. Check current state
$ git status
On branch main — nothing to commit

# 3. Stage files
$ git add index.html    # specific file
$ git add .             # all changes

# 4. Commit with message
$ git commit -m "Initial commit: add index.html"
```

Commands

git init

Start a new repo

git status

See what changed

git add .

Stage all changes

git commit

Save a snapshot

QUICK QUIZ

You run `git add index.html` then `git commit -m "add homepage"`. What happened?

A index.html was uploaded to GitHub

B A snapshot of index.html was saved to local history

C index.html was backed up to the cloud

D index.html was published online

Discuss — then we'll reveal the answer

✅ ANSWER

You run `git add index.html` then `git commit -m "add homepage"`. What happened?

A index.html was uploaded to GitHub

B A snapshot of index.html was saved to local history

C index.html was backed up to the cloud

D index.html was published online

git add stages the file; git commit saves a snapshot to your LOCAL repo history. Nothing is sent online until you git push.

Viewing History

```
$ git log
commit a3f9b12 (HEAD -> main)
Author: Juma <juma@gachon.ac.kr>
Date:   Oct 27 09:15:23 2026
    Add search feature to Book App

$ git log --oneline
a3f9b12 Add search feature to Book App
7c21e44 Fix button color on mobile

$ git diff
- old line (removed)
+ new line (added)
```

History Commands

git log

Show all commits with hash + message

git log --oneline

Compact one-line per commit

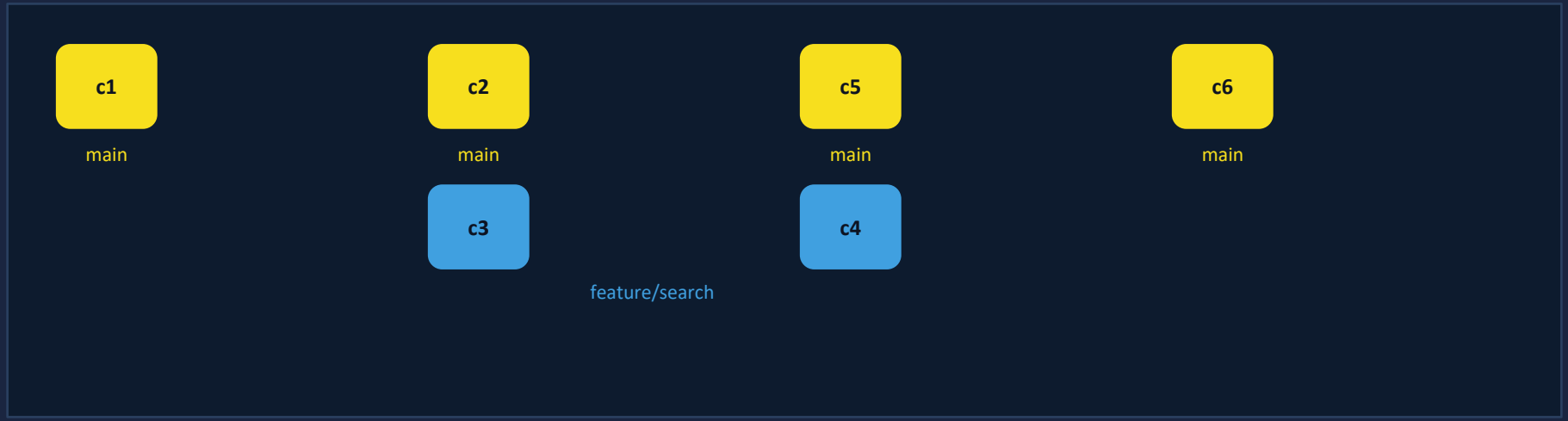
git diff

Show exactly what lines changed

Branching & Merging

Work on features in isolation — merge when ready

Branching — git branch, checkout, merge



`git checkout -b feature/search`	Create + switch to new branch
`git branch`	List all branches
`git merge feature/search`	Merge branch into main

QUICK QUIZ

What does `git checkout -b feature-login` do?

A Deletes the feature-login branch

B Merges feature-login into main

C Creates a new branch AND switches to it

D Lists all branches named feature-login

Discuss — then we'll reveal the answer

✅ ANSWER

What does `git checkout -b feature-login` do?

A Deletes the feature-login branch

B Merges feature-login into main

C Creates a new branch AND switches to it

D Lists all branches named feature-login

-b creates the branch and checkout switches to it immediately. Without -b, checkout only switches to an existing branch.

Working with Remote (GitHub)

```
# Link local repo to GitHub
$ git remote add origin https://github.com/
    username/my-project.git

# Push commits to GitHub
$ git push -u origin main

# Get teammates latest changes
$ git pull origin main

# Copy someone else's repo
$ git clone https://github.com/user/repo
```

Remote Commands

git remote add

Connect local to GitHub

git push

Upload commits to GitHub

git pull

Download + merge remote changes

git clone

Copy a remote repo locally

QUICK QUIZ

Your teammate pushed new commits. Which command downloads AND merges their changes into your branch?

A git fetch

B git clone

C git pull

D git sync

Discuss — then we'll reveal the answer

✅ ANSWER

Your teammate pushed new commits. Which command downloads AND merges their changes into your branch?

A git fetch

B git clone

C git pull

D git sync

git pull = git fetch + git merge. It downloads remote commits and applies them to your current branch. git fetch downloads but does NOT merge.

GitHub Collaboration Workflow

🍴 1

Fork

Copy the repo to your GitHub account

💻 2

Clone

Download your fork to your laptop

🌿 3

Branch

Create a feature branch

🖍️ 4

Edit & Commit

Make changes and commit them

☁️ 5

Push

Upload your branch to GitHub

🔀 6

Pull Request

Ask to merge into the original repo

Live Demo — Follow Along!

Open your terminal now

1. `mkdir book-app-git && cd book-app-git`
2. `git init`
3. `echo "Hello Git" > index.html`
4. `git status`
5. `git add index.html`
6. `git commit -m "Initial commit: add index.html"`
7. `git log --oneline`
8. `git checkout -b feature/search`
9. `echo "<input>" >> index.html`
10. `git add . && git commit -m "Add search input"`
11. `git checkout main`
12. `git merge feature/search`

Summary & Homework

git init

Create new repo

git checkout -b name

Create + switch branch

git status

Check what changed

git merge branch

Merge branch into current

git add .

Stage all changes

git push origin main

Upload to GitHub

git commit -m "msg"

Save snapshot

git pull origin main

Download from GitHub

git log --oneline

View history

HOMEWORK: Init a local Git repo for your team project. Make 3 commits (add HTML, add CSS, add JS). Push to GitHub. Share the URL on Discord by Wednesday.

QUICK QUIZ

Git is to GitHub as ____ is to ____.

A File is to folder

B Language is to website

C Local version control tool is to cloud hosting platform

D Command is to terminal

Discuss — then we'll reveal the answer

✅ ANSWER

Git is to GitHub as ____ is to ____.

A File is to folder

B Language is to website

C Local version control tool is to cloud hosting platform

D Command is to terminal

Git is the local tool that tracks changes on YOUR machine. GitHub is the cloud platform that hosts repos and enables collaboration.

Resources:

Quiz: (Git) /viewform

StudyGuide.html

VideoLecture.html

Lecture notes: Git.pdf